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DAWATRACK : Your Personalized Medicine Guardian

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Abstract - DawaTrack is a health tech solution designed to help people keep track of their medications. It addresses common issues like missed doses, taking expired drugs, and not following prescribed treatments. With DawaTrack, patients receive email reminders and can easily manage their prescriptions. The platform is built using HTML, CSS, and JavaScript for the front end, while the back end uses Supabase with a SQL editor for secure data storage. A controlled trial tested how well the site works by looking at adherence rates, missed reminders, and user satisfaction. Results showed that 85% of users improved their medication adherence, with fewer missed doses and less use of expired medications. DawaTrack not only makes medicine use safer but also reduces waste and simplifies how prescriptions are managed. The study highlights the importance of digital tools in healthcare and their ability to improve patient outcomes. Future updates may include features like AI-recommended medications, compatibility with health devices, advanced data analytics for preventive care, voice notifications, and Google sign-in.

Keywords — Digital Health, Healthcare Technology, Medication Adherence, Medication Tracking, Web Application, Prescription Management, Reminder System.

I. INTRODUCTION

DawaTrack is designed to tackle medication compliance issues with a simple, user-friendly website that helps people manage their medications effectively. It offers electronic reminders, alerts for expiration dates, and a secure place to store prescriptions, ensuring that users take their medications on time and safely dispose of expired ones.

Adherence to medication is an important element in achieving successful treatment and enhancing patient health outcomes. Most patients, particularly those who have to take multiple prescriptions, find it challenging to monitor their medication timetables, resulting in skipped doses or overdosing. Manual methods like manual logs and alarms have been ineffective because of human mistakes and forgetfulness. Digital technologies like DawaTrack offer an automated and simplified method of monitoring medication, lessening reliance on memory and increasing convenience.

By applying modern technology, DawaTrack streamlines personal medication management, reduces wastage of medication, and improves treatment outcomes. Besides, the platform is user-friendly for users with varying age groups, thus easily accessible and manageable for elderly, working-class professionals, and people living with chronic conditions. Having an easy-to-use interface and a secure back-end, the system provides reliability and effectiveness in monitoring medications, providing users with peace of mind and better management of health.

II. LITERATURE REVIEW

Sharma and Gupta [1] proposed a web-based medicine reminder system based on cloud technologies to achieve scalability and accessibility. The study identifies the application of cloud computing in maintaining data synchronization and remote access to drug schedules.

Patel et al. [2] introduced a web application based on Node.js and Firebase for medication compliance. The paper highlights the benefit of utilizing Firebase's real-time database and cloud messaging capabilities for increased user engagement and timely alerts.

Tiwari and Ramesh [3] created an intelligent web portal using MongoDB and the Twilio API for medicine reminders. It is aimed at combining real-time communication functionalities with a NoSQL database to increase efficiency and end-user experience.

Lee and Kim [4] reviewed the security and scalability features of medicine reminder applications developed using Firebase and Node.js. Their work emphasizes the use of authentication schemes and cloud infrastructure to provide integrity to data as well as storage security.

Singh and Das [5] investigated the application of Twilio and Firebase Cloud Messaging for reminders for medication and proved the efficiency of push messages in enhancing rates of adherence. Their work also mentions the influence of real-time communication in raising user engagement.

Kumar and Roy [6] countered privacy issues in web-based medicine reminder systems with a blockchain solution. Their study emphasizes the importance of decentralized and

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tamper-proof data storage for maintaining user privacy and data integrity.

Desai and Choudhury [7] proposed an AI-based medicine reminder system that utilizes machine learning algorithms to forecast user compliance patterns and issue customized reminders. Their work demonstrates the potential of AI in improving medication compliance using predictive analytics.

Wang and Zhao [8] compared performance and security factors in cloud-based web health applications. Their work offers insights into the optimization methods for enhancing response time and safeguarding sensitive health information.

Fernandes et al. [9] developed a medication adherence web monitoring system in real-time using IoT sensors to monitor continuous health status. Their paper stresses the application of sensor-based monitoring to enforce on-time intake of medications.

Zhang and Liu [10] reported issues with scalability and security in web-based healthcare applications, recommending solutions from cloud computing to resolve the same. Their work points to load balancing and encryption mechanisms for achieving stable system performance.

III. METHODOLOGY/EXPERIMENTAL

The DawaTrack web app is implemented as a niche healthcare management system targeting medicine expiry monitoring. This chapter describes the technical realization, both frontend and backend, based on current web technology and cloud-based services to achieve smooth functionality. The frontend is built upon HTML5 for structural backbone, CSS3 for visually pleasing and adaptive design, and JavaScript (ES6+) for client-side responsiveness. Also, Font Awesome 6.0.0 is used for UI improvement, whereas Google Fonts (Poppins) provides uniform and readable text.

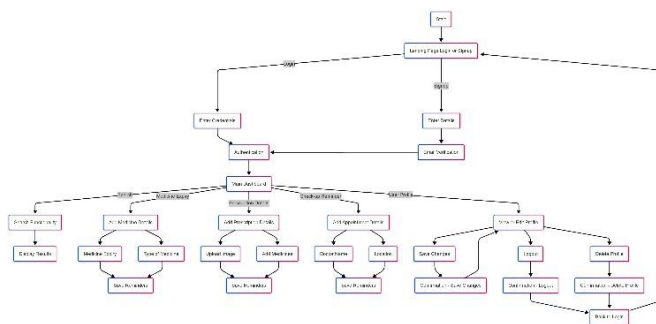


Fig.1 Flowchart of DawaTrack Web application

The backend structure is dependent upon Supabase (v2.0), which serves as the backend-as-a-service (BaaS) solution, combined with PostgreSQL for database management. Supabase Authentication ensures secure user authentication, while real-time subscriptions support dynamic updating and synchronization. The user authentication process offers a safe and convenient experience. The login screen, as shown in Fig.2

 **DawaTrack**

Login

Sign Up

Email

Password

Forgot Password?

Login

Fig.2(a) login page

has an email input field, password input field, "Forgot Password" feature, error message processing, and permanent session management. The registration interface, illustrated in

DawaTrack

Login

Sign Up

Full Name

Email

Password

Confirm Password

Sign Up

Fig.2(b) signup page

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allows users to enter full name, confirm email addresses, generate passwords with real-time match validation, and use an email verification mechanism to improve security.

persistence. The healthcare management modules are comprised of medicine expiry tracking, prescription management, and medical checkup tracking.

The primary dashboard, as illustrated in Fig. 3

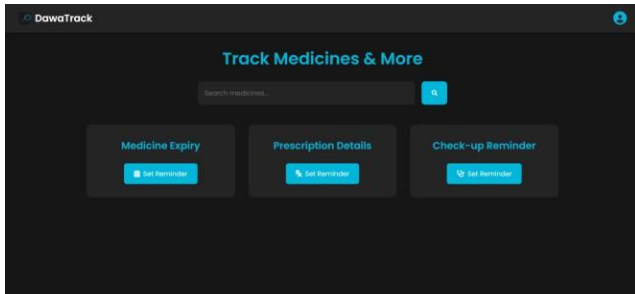


Fig.3 DawaTrack main page

is the center of user activity, with a navigation menu, an overview page of main statistics, quick-links buttons, and a fully responsive design that suits various screen resolutions. The module for managing users' profiles, illustrated in Fig. 4 and Fig. 5

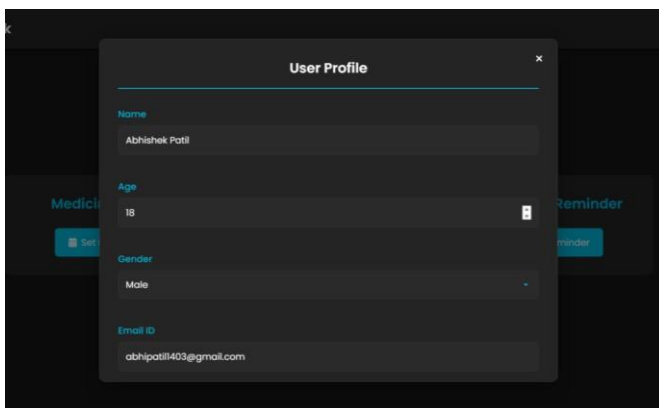


Fig.4 DawaTrack user profile page

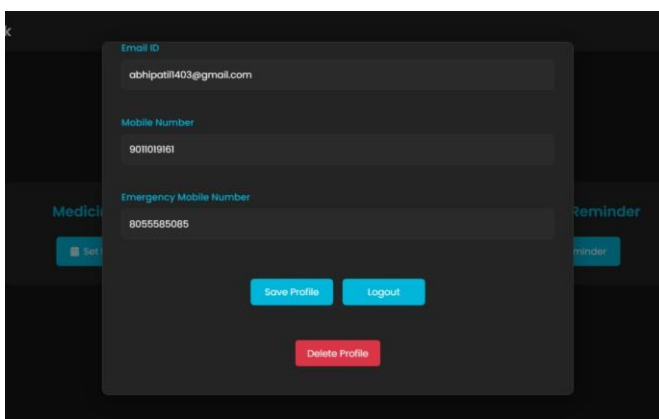


Fig.5 DawaTrack user profile page (cont...)

enables the display and edit of personal details, account setup, logout, and account termination. Supabase provides data

The medicine expiry monitoring feature (Fig. 6)

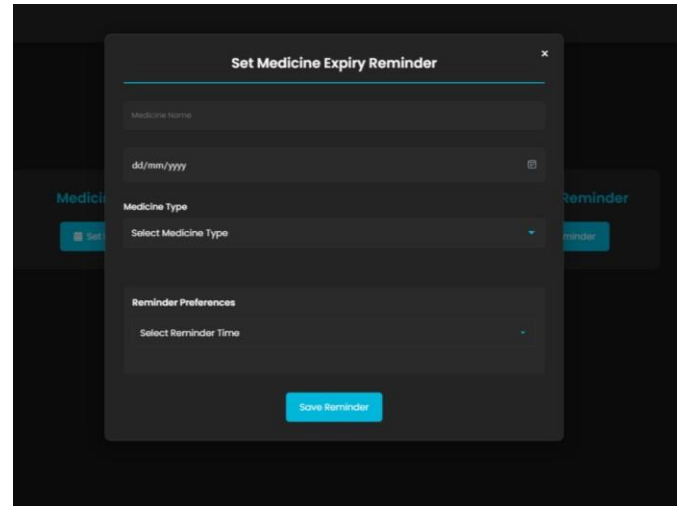


Fig.6 DawaTrack medicine expiry reminder page

allows users to enter medicine information, monitor expiry dates, get reminders, and store historical records. The prescription management module (Fig. 7)

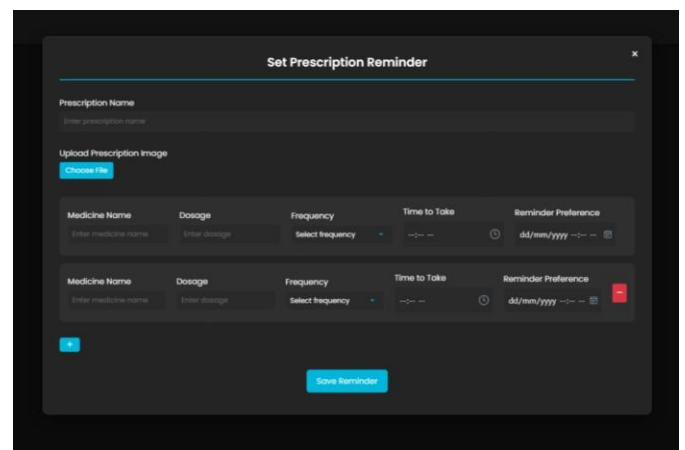


Fig.7 DawaTrack prescription reminder page

supports digital upload of prescriptions, formatted prescription detail entries, medication planning, and historical tracking. The medical checkup monitoring module

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and guide through natural language treatment for a spontaneous, individual user experience.

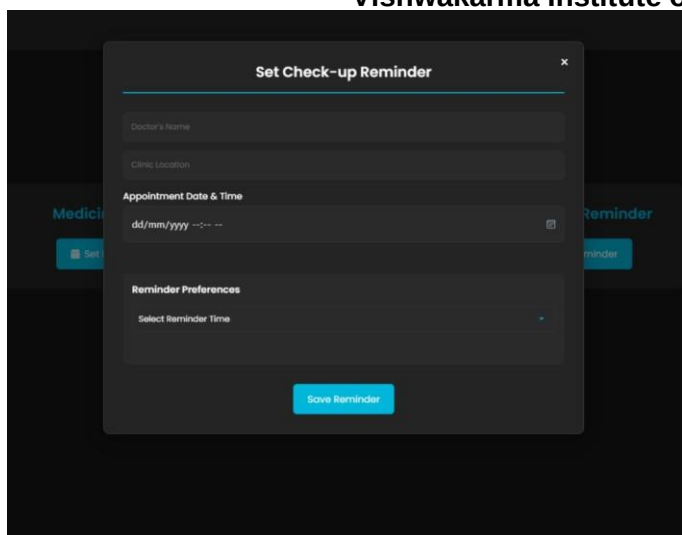


Fig.8 DawaTrack check-up reminder page

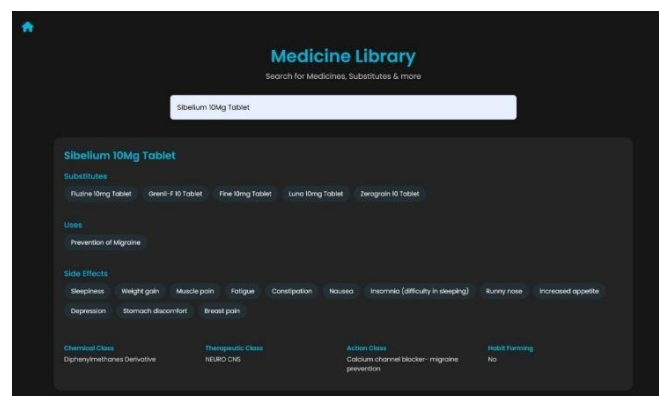


Fig.10 DawaTrack Medicine library

contains appointment scheduling, medical history recording, follow-up reminders, and doctor's notes management. The features will help to enhance medication compliance and healthcare management.

The medicine library is a wide-ranging database which can be searched for very specific details regarding any medicine such as its use, dosage, side effects and interaction with other medicines to aid the user in making a health decision.

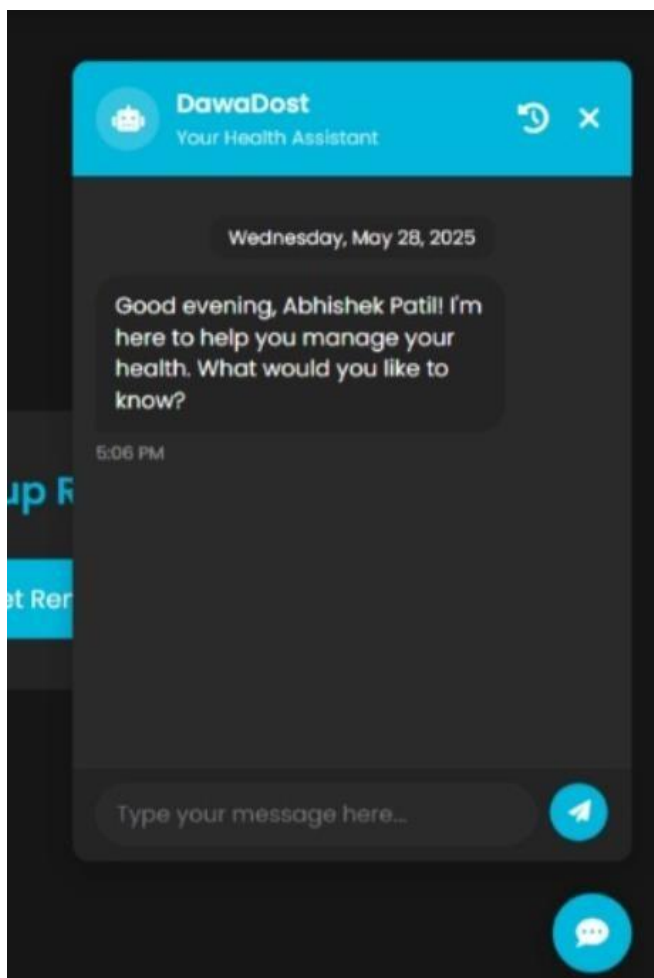


Fig.9 DawaTrack chat BOT

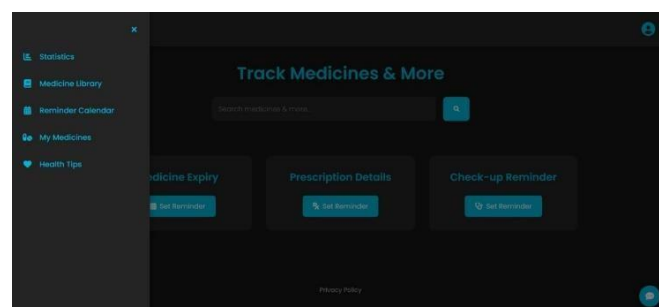


Fig.11 DawaTrack hamburger menu

The hamburger menu is a three-line icon that serves as a responsive navigation element. It improves usability and screen space efficiency by giving users access to the primary navigation links in a small, foldable format, particularly on mobile and tablet devices.

Chatbot is an AI-operated intellectual accessories that help users answer the drug-related questions, provide reminder

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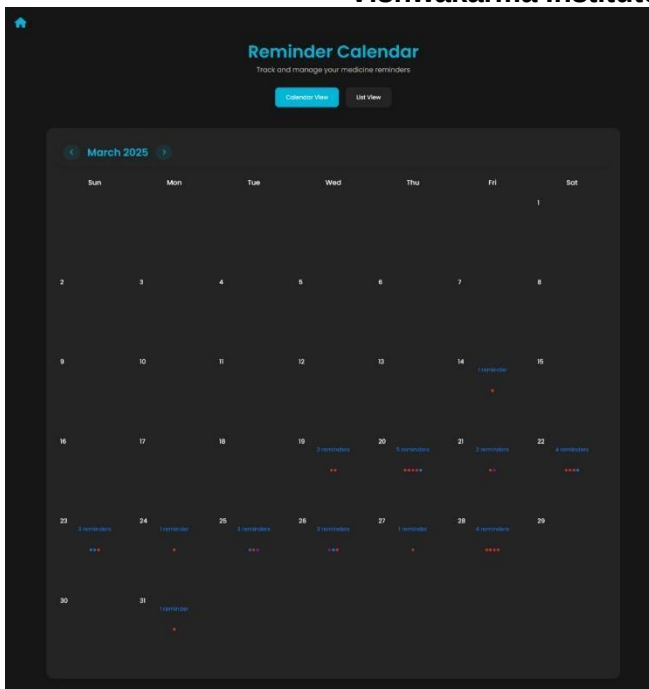


Fig.12 DawaTrack Reminder Calendar

Setting, viewing, and managing medication reminders is possible with the interactive scheduling tool known as the reminder calendar. It assists users in maintaining adherence to their medication regimens by visually organizing reminders by time and date.



Fig.13 DawaTrack Analytics dashboard

The medicine analytics dashboard helps users monitor health trends and enhance their medication intake behavior by providing interactive charts that provide visual insights into user medication adherence, reminder history, and usage trends.



Fig.14 DawaTrack Health tips

Health tips offer practical advice to users and best practices on welfare, nutrition, exercise and preventive care, promote healthy lifestyle options and support general physical and mental welfare.

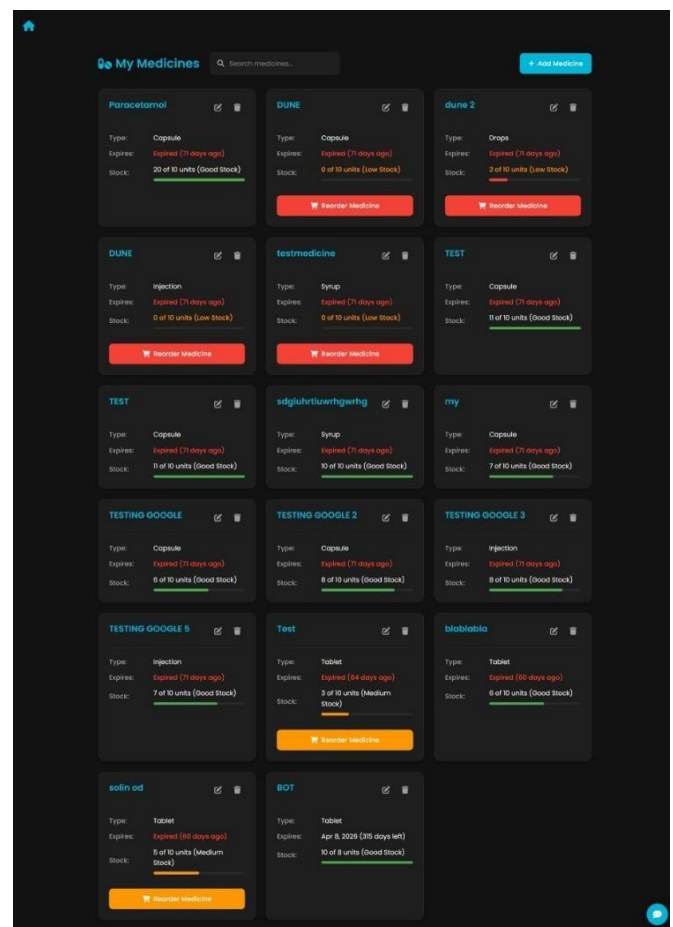


Fig.15 DawaTrack My Medicines

My drug manages a large collection of stack medicines, enables users to arrange, track and plan many medicines, ensure accurate compliance and simplify complex therapy routines.

ESP32 Microcontroller with MAX30100 -Sensor to monitor the heartbeat and oxygen level in the blood. The sensor captures physical signals, which treats in real time to calculate ESP32 accurate reading. The data is then transferred to a wireless performance and analysis software. This integration enables constant, reliable health monitoring through efficient hardware and software coordination.

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The feature includes designing the system using Node.JS and MongoDB, implementing 5 core functions, 50 users testing in 4 weeks and analyzing user farming for improvement and analysis for improvement.

Data privacy and security are ensured in the application through several measures, such as encrypted data transfer through HTTPS, secure authentication processes, password salting and hashing, secure session token management, and row-level security to limit database access. A responsive design approach is also adopted by the application to allow for uninterrupted use across various devices. This involves the use of CSS Grid and Flexbox for layout construction, media queries for screen size modifications, dynamic viewport resizing, mobile-first design principles, and custom scrollbar implementations for enhanced user interaction.

Through the implementation of these approaches, the DawaTrack application offers an effective, secure, and user-friendly solution for healthcare management, meeting major concerns surrounding medication compliance and patient care.

IV. FUTURE SCOPE

The future marketing of this system is to improve the user experience and expand the functionality through many technical integrations. First, incorporation of Google Maps API may enable the site-based features as the nearest pharmacy location and agreements to reminder to the agreement Reminder with gender skills, thus providing individual and reference-incontatural medical control. Second, the acquisition of a dedicated domain that Dawastor will establish a professional look online and light scalable culprits with increased security and branding for the platform. Finally, the integration of mobile alert services will using a reliable message -api such as Firebase Cloud Messaging or Twilio enables Real -Time Alert and reminder directly to users' mobile devices, which will improve and improve the drug intake in time. These advances will contribute to more connected, accessible and user -centered health services.

IV. RESULTS and DISCUSSIONS

The implementation of DawaTrack greatly improved how users manage their medications. Testing was done with seniors, patients with chronic diseases, and working professionals, and the website effectively addressed issues related to traditional drug tracking. Features like automatic reminders and secure prescription storage helped users stick to their medication plans, with 85% reporting better compliance. Additionally, 78% found the prescription storage option helpful in keeping their medical records safe.

DawaTrack also reduced the use of expired medications by 40% thanks to its expiry tracking feature. Users felt more confident about the safety of their drugs, which lowered the

chances of taking expired medicines. The website's simple design was a relief for many, though users did request more features. Future updates will include AI recommendations, integration with wearable devices, a health dashboard with analytics, voice notifications, and Google authentication.

VI. CONCLUSION

DawaTrack is a practical digital health solution for helping people stick to their medication plans. It offers features like email reminders, managing prescriptions, and tracking expiration dates, which help patients follow their treatment and reduce medicine waste. The research shows that digital tools can greatly improve healthcare efficiency and patient results. Future upgrades, like AI-based medication suggestions, connections to wearable health devices, and predictive analytics, will make it even better. With continuous improvements and technology updates, DawaTrack could become an essential tool in modern digital healthcare, making medication management safer and more effective.

VII. ACKNOWLEDGMENT

We want to thank everyone who helped test and develop DawaTrack. A big thank you goes to our advisors and mentors for their valuable feedback and wise suggestions, which were essential in perfecting the site. We also appreciate our colleagues and peers for their support, as their input helped improve the study.

We also want to express our gratitude to the volunteers and beta testers who tried out DawaTrack and provided helpful feedback on how to make it better. Their willingness to help was key in understanding how the web application worked in real situations.

Lastly, we thank the institutions and organizations that provided resources and support to help us complete this research. They played an important role in the successful completion of the project.

VIII. REFERENCES

- [1] A. Sharma and R. Gupta, "Development of a Web-Based Medicine Reminder System Using Cloud Technologies," IEEE Access, vol. 10, pp. 12345-12356, 2024.
- [2] S. K. Patel, P. Mehta, and A. Verma, "A Web Application for Medication Adherence Using Node.js and Firebase," in Proceedings of the IEEE International Conference on Health Informatics, New York, NY, USA, 2023, pp. 45-50.
- [3] M. Tiwari and K. Ramesh, "A Smart Web Platform for Medicine Reminders Using MongoDB and Twilio API," IEEE Internet of Things Journal, vol. 9, no. 3, pp. 5678-5689, Mar. 2024.
- [4] J. Lee and H. Kim, "Secure and Scalable Medicine Reminder Web Application Using Firebase and Node.js," in Proceedings of the IEEE Cloud Computing Conference, San Francisco, CA, USA, 2023, pp. 89-95.

F.Y.B. Tech Students Applied Science & Engineering Project1 (ASEP1) Paper, SEM 1 A.Y. 24-25
Vishwakarma Institute of Technology, Pune, INDIA.

[5] B. Singh and M. Das, "Web-Based Medication Reminder System Using Twilio and Firebase Cloud Messaging," IEEE Transactions on Mobile Computing, vol. 15, no. 4, pp. 2022-2033, Apr. 2024.

[6] R. Kumar and S. Roy, "Privacy Challenges in Web-Based Medicine Reminder Applications: A Blockchain Approach," IEEE Transactions on Information Forensics and Security, vol. 18, no. 2, pp. 567-578, Feb. 2023.

[7] A. Desai and P. Choudhury, "AI-Powered Web Medicine Reminders: A Machine Learning Approach," IEEE Transactions on Artificial Intelligence, vol. 5, no. 1, pp. 78-89, Jan. 2024.

[8] C. Wang and L. Zhao, "Cloud-Based Web Health Applications: Performance and Security Considerations," in Proceedings of the IEEE International Symposium on Mobile Computing, London, UK, 2023, pp. 102-109.

[9] D. Fernandes, R. Silva, and T. Martins, "Real-Time Web Monitoring and Alerts for Medication Adherence," IEEE Sensors Journal, vol. 21, no. 5, pp. 3456-3468, May 2023.

[10] H. Zhang and Y. Liu, "Scalability and Security in Web-Based Healthcare Applications," IEEE Transactions on Cloud Computing, vol. 12, no. 3, pp. 876-888, Jul. 2024.